

Plan

- Menti on 2D models 5-10 min
- Some common pitfalls 5 min
- Lecture on stability 15 min
- Break 10 min
- Exercises 1 hour

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Some misunderstandings

Cylindrical 2D

$$\frac{\partial C_A}{\partial t} = D \left(\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial C_A}{\partial r} \right) + \frac{\partial^2 C_A}{\partial z^2} \right) + R_A$$

Spherical 1D

$$\frac{\partial C_A}{\partial t} = D \left(\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial C_A}{\partial r} \right) \right) + R_A$$

r is NOT constant. Need to use product rule of differentiation

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial C_A}{\partial r} \right) = \frac{1}{r^2} \left(r^2 \frac{\partial^2 C_A}{\partial r^2} + 2r \frac{\partial C_A}{\partial r} \right) = \frac{\partial^2 C_A}{\partial r^2} + \frac{2}{r} \frac{\partial C_A}{\partial r}$$

What is advection!?

$$\frac{\partial C_A}{\partial t} + v_x \frac{\partial C_A}{\partial x} + v_y \frac{\partial C_A}{\partial y} + v_z \frac{\partial C_A}{\partial z} = D_{AB} \left(\frac{\partial^2 C_A}{\partial x^2} + \frac{\partial^2 C_A}{\partial y^2} + \frac{\partial^2 C_A}{\partial z^2} \right) + R_A.$$

Movement of the bulk. Look for "velocity, flow" in problem text.

Can advection occur in a solid material?

A mixer is stirring a reactor – is it advection?

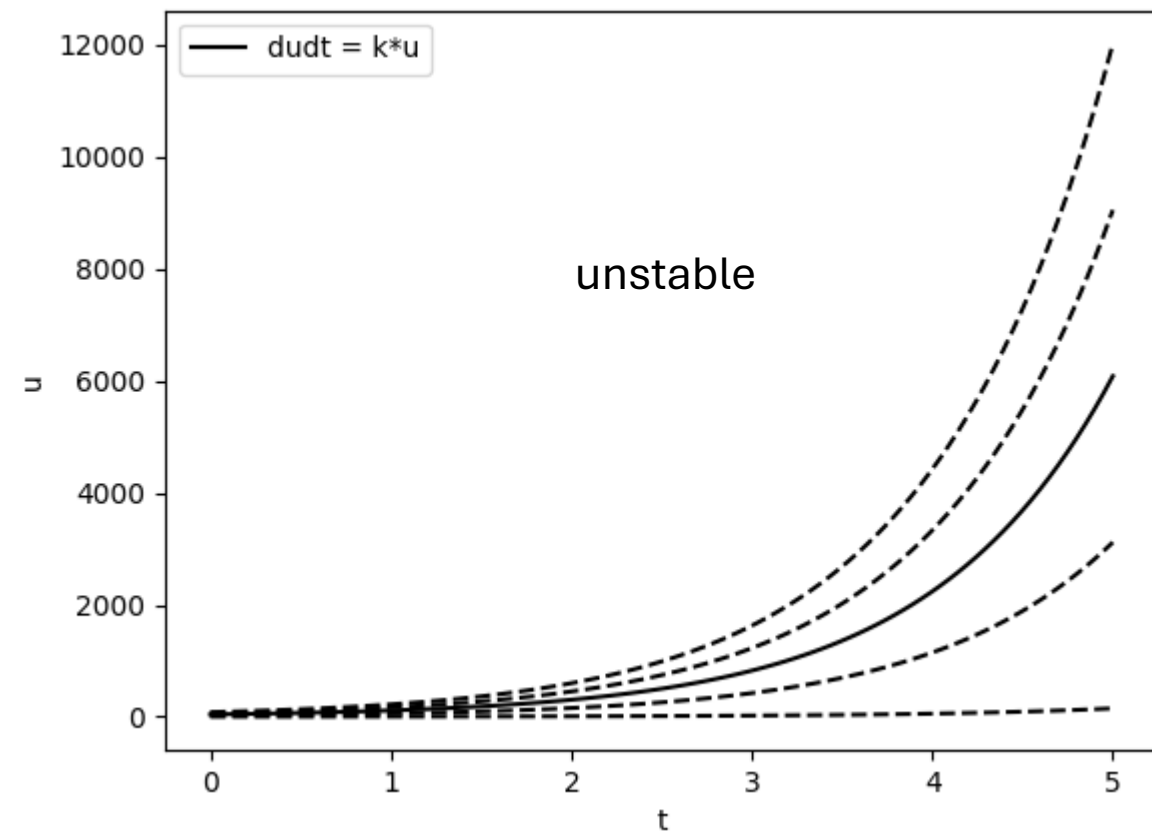
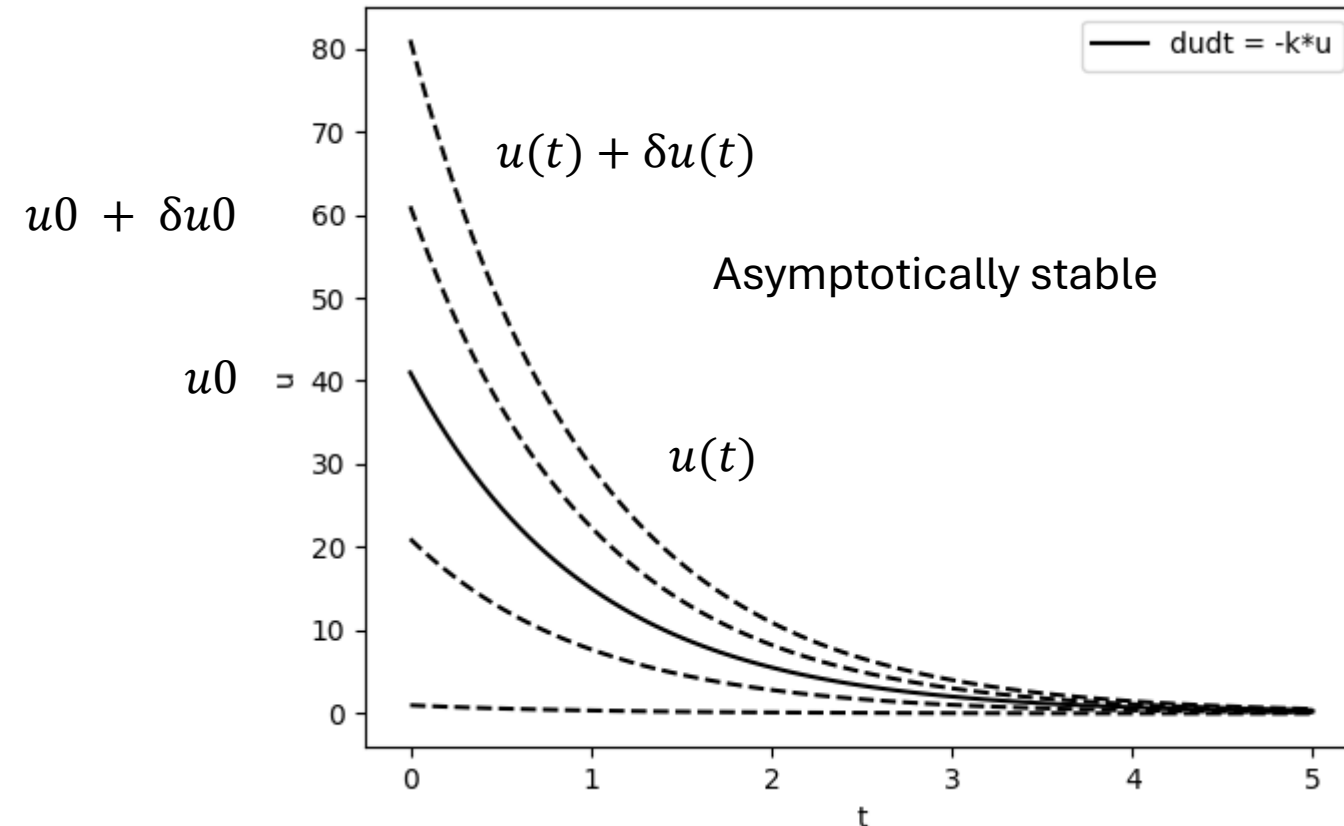
Stability and stiffness

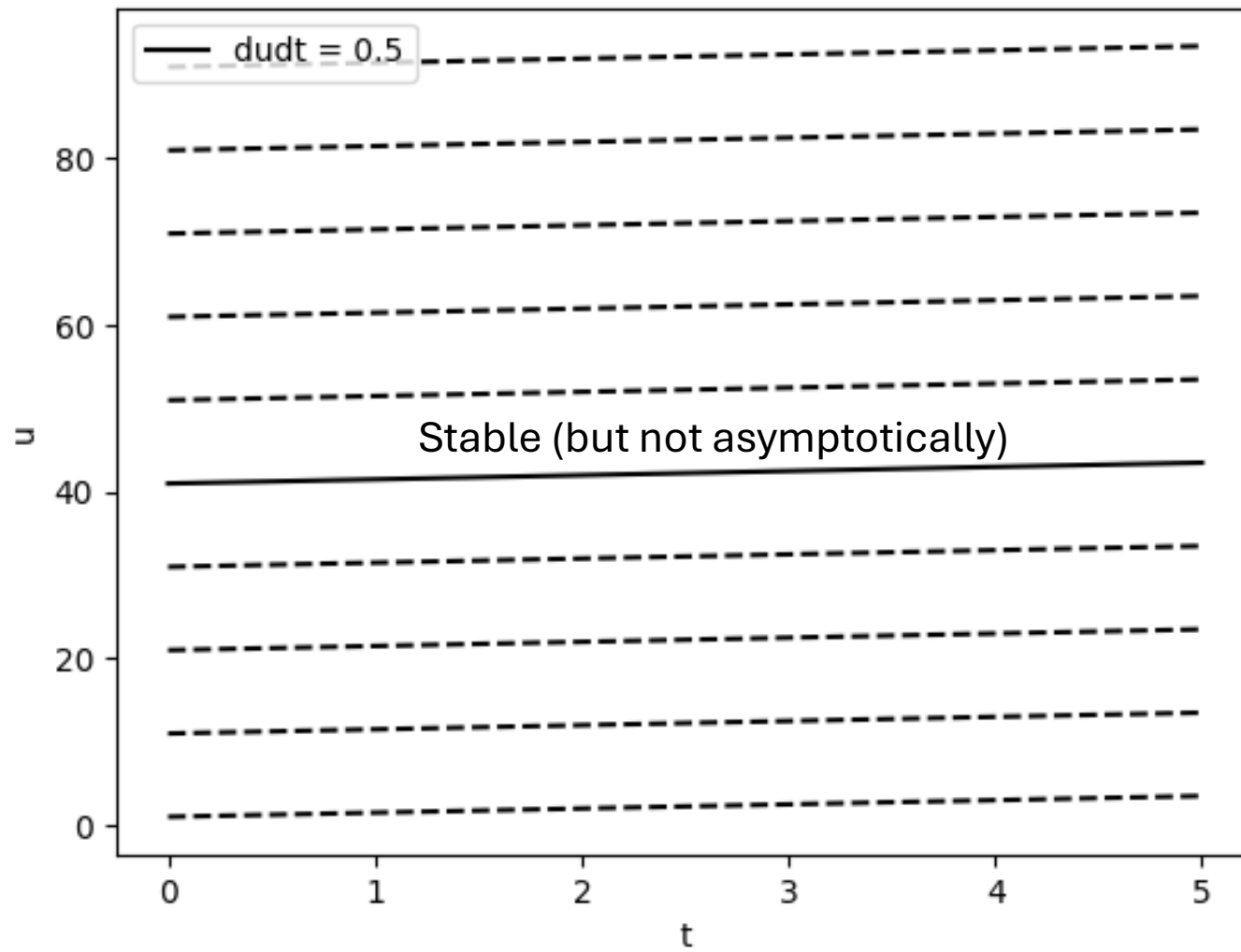
Modelling Fall 2025

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Stability of an ODE/ODE system

A solution is stable if every solution, which starts sufficiently close to the solution remains close to the solution for all $t > t_0$





ODE system

If one of many ODEs are unstable, then the whole ODE system is unstable!

For a linear ODE system with constant coefficients (LCC).

$$\frac{d\mathbf{u}}{dt} = A\mathbf{u}$$

Calculate eigenvalues (λ 's) of A then:

If all real parts of λ 's < 0 , the system is stable

If the real parts of any λ 's > 0 , the system is unstable

If any real parts of λ 's $= 0 \rightarrow$ marginal case:

- Purely imaginary, then Lyapunov stable, (oscillatory)
- Imaginary part also $= 0$, we will not go that way..

Python call is `np.linalg.eigvals(A)`

How to get A in $\frac{d\mathbf{u}}{dt} = A\mathbf{u}$ from multiple ODEs in the LCC case?

ODE system

$$\frac{du_1}{dt} = -u_1 + 3u_2$$

$$\frac{du_2}{dt} = 0.5u_1 - 2u_2$$

Columns are coefficient in front of

u_1

u_2



$$\frac{d\mathbf{u}}{dt} = \begin{bmatrix} -1 & 3 \\ 0.5 & -2 \end{bmatrix} \mathbf{u}$$

$\leftarrow \frac{du_1}{dt}$

$\leftarrow \frac{du_2}{dt}$

Rows are the different ODEs

$$A = \begin{bmatrix} -1 & 3 \\ 0.5 & -2 \end{bmatrix}$$

Eigenvalues $\leftarrow$ `np.linalg.eigvals(A)`

Stability in non-linear ODE systems

Stability of nonlinear ODE is not straight forward and

$$\frac{d\mathbf{u}}{dt} = A\mathbf{u}$$

Is not possible with constant coefficients in A. So we instead have some nonlinear function vector, $\mathbf{f}$.

$$\frac{d\mathbf{u}}{dt} = \mathbf{f}(\mathbf{u})$$

We still want to know if $\mathbf{u}(t) + \delta\mathbf{u}(t)$ stays close to $\mathbf{u}(t)$ (condition for stability)

Linearization with taylor expansion will get us to

$$\frac{d\mathbf{u}}{dt} + \frac{d\delta\mathbf{u}}{dt} = \mathbf{f}(\mathbf{u}) + \frac{\partial \mathbf{f}}{\partial \mathbf{u}} \delta\mathbf{u} + O(|\delta\mathbf{u}|^2)$$

$$\frac{\partial \mathbf{f}}{\partial \mathbf{u}} = J(\mathbf{u}) \text{ where } \mathbf{f} = \frac{d\mathbf{u}}{dt}$$

J is the **jacobian**

Stability of the nonlinear ODE system around critical points

Critical points/steady state/stationary solutions is where

$$\frac{d\mathbf{u}}{dt} = \mathbf{f}(\mathbf{u}) = 0$$

Is the critical points stable or unstable?

1. Find the solution (u^*) to $\frac{d\mathbf{u}}{dt} = \mathbf{f}(\mathbf{u}) = 0$ (use fsolve)
2. Evaluate the eigenvalues of $J(u^*)$ to assess stability as for LCC systems, where u^* is a critical point.

(use numdifftools.Jacobian)

$$J(\mathbf{u}) = \frac{\partial \mathbf{f}}{\partial \mathbf{u}} \quad \text{where } \mathbf{f} = \frac{d\mathbf{u}}{dt}$$

$$\frac{du_1}{dt} = f_1(\mathbf{u}) = u_1 - 0.1 u_1 u_2$$

$$\frac{du_2}{dt} = f_2(\mathbf{u}) = 0.075 u_1 u_2 - 1.5 u_2$$

$$J(\mathbf{u}) = \begin{bmatrix} \frac{\partial f_1}{\partial u_1} & \frac{\partial f_1}{\partial u_2} \\ \frac{\partial f_2}{\partial u_1} & \frac{\partial f_2}{\partial u_2} \end{bmatrix} = \begin{bmatrix} 1 - 0.1u_2 & -0.1u_1 \\ 0.075u_2 & 0.075u_1 - 1.5 \end{bmatrix}$$

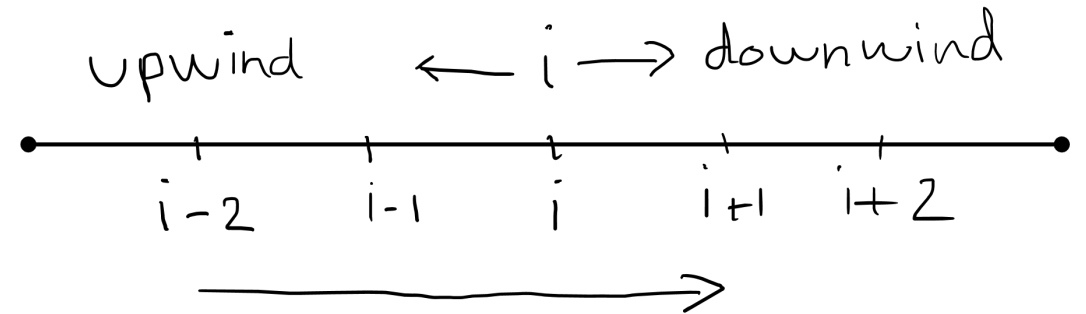
Demonstration python

Stiffness in ODE solutions

- Stable ODE system (from analytical point of view) can still result in unstable solution if care is not taken!
- Stability of solution depends on both method of discretization and time integration method!
- Some solvers with adaptive stepsize → avoid stability issue, but increase time to solve it
- Stiffness ratio. Largest/smallest eigenvalue. Rule of thumb: above 1000 it is stiff

PDE type	Time integration	Spatial order / discretization (x)	Stability criterion	Stable?
Diffusion	Explicit Euler	2nd-order central	$dt \leq dx^2 / (2 D)$	Yes, if criterion met
Diffusion	Implicit Euler	2nd-order central	Unconditionally stable	Yes
Diffusion	RK4 (explicit)	2nd-order central	$dt \leq 0.696 dx^2 / D$	Yes, if criterion met
Advection	Explicit Euler	1st-order upwind (backward for $v > 0$)	$dt \leq dx / \text{abs}(v)$	Yes, if criterion met
Advection	Explicit Euler	2nd-order central	Unconditionally unstable	No
Advection	Implicit Euler	1st-order upwind or 2nd-order central	Unconditionally stable	Yes
Advection	RK4 (explicit)	1st-order upwind	$dt \leq 2 dx / \text{abs}(v)$ (approx.)	Yes, if criterion met
Advection	RK4 (explicit)	2nd-order central	$dt \leq 2.828 dx / \text{abs}(v)$ (approx.)	Yes, if criterion met
Reaction	Explicit Euler	—	$dt \leq 2 / \text{abs}(\lambda)$	Yes, if criterion met
Reaction	Implicit Euler	—	Unconditionally stable	Yes
Reaction	RK4 (explicit)	—	$dt \leq 2.785 / \text{abs}(\lambda)$	Yes, if criterion met
Advection–Diffusion	Explicit Euler	1st-order upwind + 2nd-order central	$dt \leq \min(dx/\text{abs}(v), dx^2/(2 D))$	Yes, if criterion met
Advection–Diffusion	Explicit Euler	2nd-order central + 2nd-order central	$dt \leq \min(0.5 dx/\text{abs}(v), dx^2/(2 D))$	Conditionally, small dt
Advection–Diffusion	Implicit Euler	1st- or 2nd-order + 2nd-order central	Unconditionally stable	Yes
Advection–Diffusion	RK4 (explicit)	1st-order upwind + 2nd-order central	$dt \leq \min(2 dx/\text{abs}(v), 0.696 dx^2/D)$	Yes, if criterion met
Advection–Diffusion	RK4 (explicit)	2nd-order central + 2nd-order central	$dt \leq \min(2.828 dx/\text{abs}(v), 0.696 dx^2/D)$	Yes, if criterion met

Upwind: when using finite difference, you have to use the method that includes the upwind state variable.



$$\frac{C_i - C_{i-1}}{\Delta x} = \text{backward diff}$$